Standard Operating Procedure: Resolving Huawei Access Node Optical Transport Failure ALARM_2193 in YEL-ACC-52 at Yelahanka Sub-station

Node: YEL-ACC-52 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_2193

Description

The Huawei Access Node Optical Transport Failure in YEL-ACC-52 at Yelahanka Sub-station may occur due to the degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss in the optical transport domain. This failure domain is specific to the Access Node role of Huawei equipment, which utilizes WDM (Wavelength Division Multiplexing) technology to transport multiple optical signals over a single fiber. The WDM system requires accurate alignment and synchronization of optical signals to prevent interference and ensure reliable transmission. A degradation of SFP/XFP transceivers, fiber cuts, or Rx/Tx power loss can disrupt this alignment, resulting in signal degradation and triggering the ALARM_2193 alarm. It is essential to identify and resolve these issues promptly to prevent network disruptions and maintain high-quality signal transmission.

Pre-requisite Checks

- Verify that the Access Node YEL-ACC-52 is accessible via the Huawei CLI.

- Confirm that the redundant link (if installed) is functioning correctly.

- Check the environmental conditions at Yelahanka Sub-station, including temperature, humidity, and physical security.

- Ensure that the remote hands team is dispatched to Yelahanka Sub-station for assistance if required.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-channel current-status, Expected Output: Optical channel current-status information, Description: This command displays the current status of the optical channels, including the received optical power, transmit optical power, and signal quality. This information can help identify potential issues with the SFP/XFP transceivers or fiber cuts.
```

```
Command: display optical-channel tx-power, Expected Output: Optical channel transmit power information, Description: This command displays the transmit power of the optical channels, which can help identify potential issues with the SFP/XFP transceivers or fiber cuts.

Command: display optical-channel rx-power, Expected Output: Optical channel received power information, Description: This command displays the received power of the optical channels, which can help identify potential issues with the SFP/XFP transceivers or fiber cuts.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceivers, Command: replace optical-channel transceiver , Description: Replace the SFP/XFP transceivers with new ones to resolve the issue. Ensure that the new transceivers are compatible with the Huawei equipment and installed correctly.

Step 2: Step: Repair or replace the fiber, Command: repair optical-fiber , Description: Repair or replace the fiber to resolve the issue. Ensure that the new fiber is compatible with the Huawei equipment and installed correctly.

Step 3: Step: Adjust the Rx/Tx power settings, Command: adjust optical-channel power , Description: Adjust the Rx/Tx power settings to resolve the issue. Ensure that the new power settings are compatible with the Huawei equipment and do not cause any other issues.

Post-Check Verification

```
Command: display optical-channel current-status, Expected Output: Optical channel current-status information, Description: Verify that the optical channel current-status information is displayed correctly after the resolution steps.

Command: display optical-channel tx-power, Expected Output: Optical channel transmit power information, Description: Verify that the optical channel transmit power information is displayed correctly after the resolution steps.

Command: display optical-channel rx-power, Expected Output: Optical channel received power information, Description: Verify that the optical channel received power information is displayed correctly after the resolution steps.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceivers, Command: restore optical-channel transceiver , Description: Restore the original SFP/XFP transceivers if the replacement transceivers do not resolve the issue.

- Step: Restore the original fiber, Command: restore optical-fiber , Description: Restore the original fiber if the replacement fiber does not resolve the issue.

- Step: Restore the original Rx/Tx power settings, Command: restore optical-channel power , Description: Restore the original Rx/Tx power settings if the adjusted power settings do not resolve the issue.